

Multi-task Learning for Stress Recognition

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ABSTRACT

Different people experiencing the same stressor can have different psycho-physiological reactions. Such differences happen even when reporting the same affective state. While this mismatch causes a drop in affect recognition performance, researchers have shown that this same psycho-physiological uniqueness can be utilized to develop sensor-based biometric recognition systems. These findings open up the possibility to exploit the benefits of the psycho-physiological uniqueness observed in biometric recognition systems to improve affect machine learning models. In this paper, we explore the joint learning of a model for stress recognition and user identification to improve the performance of the first task. The joint model is learned using Multi-task Learning (MTL) neural network. We show that MTL achieves a 6.5 percentage points improvement compared to single-task models (from 86.1 to 92.6 F1-score). These results inform the design and development of stress recognition systems.

CCS CONCEPTS

• Human-centered computing → Ubiquitous and mobile computing systems and tools.

KEYWORDS

User Identification, Stress Recognition, Machine Learning, Wearable Sensors

ACM Reference Format:

Alessandro Pogliaghi, Elena Di Lascio, Shkurta Gashi, Emanuela Piciuccio, Silvia Santini, and Martin Gjoreski. 2022. Multi-task Learning for Stress Recognition. In *Proceedings of the 2022 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp/ISWC '22 Adjunct)*, September 11–15, 2022, Cambridge, United Kingdom. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3544793.3563404>

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UbiComp/ISWC '22 Adjunct, September 11–15, 2022, Cambridge, United Kingdom

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ACM ISBN 978-1-4503-9423-9/22/09...\$15.00

<https://doi.org/10.1145/3544793.3563404>

1 INTRODUCTION

Human affect is part of natural everyday communication. Affect can be expressed in various ways, such as, e.g., verbally or through facial expressions and hand gestures. Research shows that affective states can have a crucial role in the decision-making process, which is strongly related to tendencies to perform actions. These capabilities and the ability to understand or feel and react to affective states of other human beings are referred to as the *"emotional intelligence"* [14]. Including affect in the human-machines communication is defined as *"affective computing"* and *affect recognition* is an important problem in affective computing. One of the applications of affect recognition systems is to forge machines that are capable of detecting humans under stress conditions, with the goal of improving the interaction between humans and machines [21].

The affective states of human beings can be recognized using sensing devices. Various approaches have been established exploiting audio-visual expressions or the analysis of physiological signals and body movements [4]. Regarding sensor technologies, though, the commonly utilized, e.g., the electromyography sensors, were bulky and expensive lab-specialized devices. These devices were intrusive with respect to the end-users, not affording extensive usage in everyday settings [2]. The situation changed with the breakthroughs in hardware and software engineering, which led to the availability, first for professionals and academic purposes and then for public adoption, of wearable devices. The major contribution of wearable devices equipped with multiple sensors is the capability of capturing behavioral and physiological traits in a non-invasive manner.

In the research community, there are still open challenges that need to be tackled to make stress recognition ready for real-life deployment. A major issue is the uniqueness and interpersonal variability of the physiological responses of humans to stimuli [19]. Different people experiencing the same stressor, or even reporting the same affective state, can have different psycho-physiological reactions. This mismatch might cause a drop in the performance of stress recognition models. On the other hand, this psycho-physiological uniqueness has been used recently to develop wearable biometric recognition systems with encouraging results [10, 22].

The main goal of this article is to investigate whether we can exploit the benefits of the psycho-physiological uniqueness observed in biometric recognition systems to improve affect recognition machine learning models. To do so, we use Multi-task learning (MTL)

to jointly recognize users' stress levels and identity. The MTL paradigm could allow exploiting the commonalities and differences across these two tasks to improve the performance of the main task. A model able to jointly recognize user identity and affective state would enable the delivery of appropriate and personalized intervention strategies. MTL for jointly recognizing users and performing other classification tasks has been previously studied in other domains, with activity recognition from wearable accelerometers being the closest one to our work [9, 24]. To the best of our knowledge, this is the first study exploring the conjunction of physiological data from wearable sensors and user identification for stress recognition via MTL.

2 BACKGROUND

2.1 Stress recognition

Stress is a process by which a stimulus elicits an emotional, behavioral and/or physiological response, which is conditioned by an individual's personal, biological and cultural context [16]. The autonomic nervous system (ANS) system generates a response to a stress process. The ANS is the primary mechanism in control of the fight-or-flight response, during which a mixture of hormones (like adrenaline) is released, leading, for example, to increased heart rate and sweating rate. These physiological changes prepare the organism for a faster response to the stress stimulus [17]. The human body can overreact to stressors that are not life-threatening (e.g. work pressure). Phobias are an example of how the fight-or-flight response might be falsely triggered in the face of a perceived threat. The ANS is composed of two subsystems, the Sympathetic Nervous System (SNS) and the Parasympathetic Nervous System (PNS). The SNS is responsible for innervating all vascular smooth muscle and sweat glands and its stimulation results in accelerated heart rate and increased blood pressure, among others. The PNS is responsible for the body's rest and digestion response when relaxed, during resting, or feeding. It undoes the damage caused by the SNS activation during a stressful situation. Stimulation of the PNS results in decreased heart rate and blood pressure, among other effects. The interaction between PNS and SNS can be quantified by analyzing changes in the related physiological signals, e.g., via the Heart Rate Variability. Additionally, the Electrodermal Activity (EDA) is used to measure changes in the SNS subsystem [13].

2.2 User recognition

The task of user recognition can be seen as a pattern recognition problem in which a user provides a set of physiological and/or behavioral characteristics to match a previously registered signature that allows the system to verify the subject [6]. This task takes advantage of the fact that humans have natural traits that are inherently unique for each individual. Typical applications involve controlling users' access to resources, e.g., transportation vehicles, computer systems, or a building. In such systems, the users need to actively present themselves to the system to be recognized. Users' identities were mapped via a pre-defined ID label.

2.3 Multi-task Learning

Multi-task learning (MTL) [7] is a specific *machine learning* (ML) paradigm in which a model is jointly trained for multiple tasks (the

types of prediction or inference based on the problem at hand), instead of training on a single task, as in the more traditional ML paradigm of single-task learning. The architectural structure of such models is composed of shared layers and task-specific layers. The shared layers allow the model to be trained on multiple tasks, and thus to share common information about the two tasks. This common information should help the model to generalize better compared to single-task models. To build such a model, an inner relationship between the tasks should be present [30].

MTL models are typically performed via deep learning (DL) models [20]. MTL models allow the DL network to learn more generalized representations by learning different (but related) tasks jointly, resulting in a performance boost and a more robust model ready to be deployed in the wild. The effectiveness of MTL models can be seen in the fact that it utilizes more data from different learning tasks than single-task learning. Thanks to the larger volume of data, MTL models can obtain improved knowledge sharing among the different tasks, thus resulting in better performance and a lower risk of overfitting each of the learning tasks. Some fields in which MTL has been used successfully are natural language processing (NLP), computer vision, and wearable computing, where the model architecture proposed showed that the shared network architecture allowed the models to be more effective [24].

3 DATASET

To investigate the impact of joint learning of stress recognition and user identification model using multi-task learning, we use the publicly available *WEearable Stress and Affect Detection (WESAD)* dataset [27]. *WESAD* includes data from 15 subjects (12 males and 3 females) recorded in a laboratory setting, each of the subjects involved in the study experienced four conditions: *Baseline* – reading task; *Amusement* – watching funny videos; *Stress* – exposure to the TSST [18]; and *Meditation*, de-excite procedure for bringing back the subjects to a more neutral affective state. The Baseline condition was recorded for 20 minutes, Amusement for 392 seconds (~ 6.5 minutes), Stress for 10 minutes, and lastly Meditation for 7 minutes. The *WESAD* dataset contains motion and physiological data recorded from a wrist and chest-worn device. For this particular article, the focus was on the *EDA* and *BVP* gathered via the wrist-worn device, an Empatica E4 [12]. The Empatica E4 records *EDA* and *BVP* at 4Hz and 64Hz sampling frequencies respectively.

4 METHOD

Fig. 1 shows the ML pipeline we designed. The pipeline consists of five major steps: *data acquisition*, *pre-processing*, *segmentation*, *feature extraction* and *feature cleaning*. More details of each step are presented in following sections.

4.1 Pre-processing and segmentation

Physiological signals collected using wearable devices can be affected by artifacts that can hamper the reliability of the results. Artifacts are caused by the misplacement of the device and movement. To reduce them, we applied a filtering procedure to the raw *EDA* and *BVP* data. Specifically, we used a first-order Butterworth low-pass filter with a cutoff of 5Hz for *BVP* and 0.4Hz for *EDA*. We then segmented the filtered sensor data using windows of 60

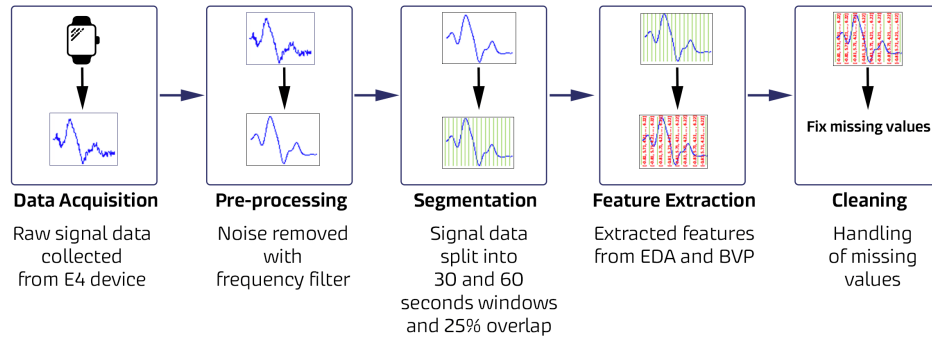


Figure 1: Architectural Overview of the Data pipeline

seconds with a 25% overlap. We then extracted features from the derived segments.

4.2 Feature extraction

We extracted 77 features from EDA and BVP used in the literature [25], to provide a high-level representation of the signals for *stress recognition* and *user identification*. We used such features in input to machine learning models.

The EDA signal consists of two components, *phasic*, a fast-changing signal related to the momentary stimulus, and *tonic* or slow-changing [3]. The EDA is characterized by peaks also known as *skin conductance responses (SCRs)* [3]. We decomposed the EDA using the *cvxEDA* algorithm presented in [15], which uses methods of convex optimization to describe the activity of the autonomic nervous system in response to strong affective stimulation. We extracted a total of 50 features presented in the literature from the EDA and its components using the publicly available tool *EDAExplorer* [1]. A summary of the features is presented in Table 1.

We derived 27 features from BVP to characterize cardiovascular activity. We extracted features in the time and frequency domain using the *HeartPy* tool [29] as well as statistical features (e.g., mean, standard deviation), as a common practice in the literature [11]. Table 2 presents the list of extracted features from the BVP.

4.3 Imputation and normalization

During the feature extraction procedure, it is possible that an accurate computation of some features may not be possible due to missing, noisy or invalid data originating from the starting dataset. For some features, the number of missing values amounted to 47.51% of the total available data points. We imputed the missing values using an iterative approach based on multivariate regression. This approach involves defining a regression model (Bayesian ridge regressor) to predict each of the missing features as a function of all the other features. This method has been reported to produce more accurate estimations compared to the use of constants or statistical values [5]. We then normalized the features to reduce the data-related variability between subjects and to scale the numerical values to a pre-defined range. This allows the ML models to more quickly learn the optimal parameters for each input. In this paper,

Table 1: EDA features

Features	Explanation
<i>peaks</i>	# of peaks
<i>rise_time</i>	avg rise time of the peaks
<i>max_deriv</i>	avg. value of the max derivative
<i>amp</i>	avg. amplitude of the peaks
<i>decay_time</i>	average decay time of the peaks
<i>SCR_width</i>	average width of the peak for Skin Conductance Response
<i>auc</i>	average area under the curve for the peak
<i>median</i>	value separating the higher half from the lower half
<i>mean</i>	arithmetic mean
<i>std</i>	standard deviation
<i>var</i>	statistical variance
<i>slope</i>	slope of the regression line
<i>min</i>	minimum value of the signal
<i>max</i>	maximum value of the signal
<i>fdmean</i>	mean value of the gradient on the EDA signal
<i>fdstd</i>	standard deviation value of the gradient on the signal
<i>drange</i>	ratio between the largest and smallest values

we used a Min-Max normalization procedure on a *per-feature basis*, scaling the extracted features values between 0 and 1.

4.4 Classification Pipeline

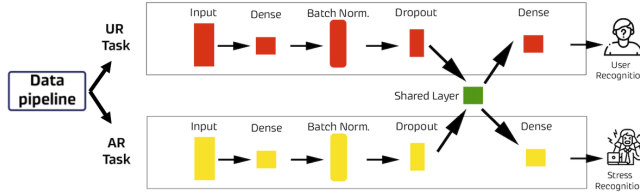
In this work, we experimented with a MTL strategy implemented using DNNs to jointly solve the stress recognition and user identification tasks. We compare the performance of this model with a single-task strategy and shallow classifiers. We describe below the models and the evaluation procedure we used.

We considered two widely used [13, 27] machine learning algorithms: *Random Forest (RF)* and *Extreme Gradient Boosting (XGBoost)* as baselines. We chose these algorithms because they do not require extensive hyperparameter tuning and are robust against noisy data.

The MTL configuration is built using a Feed-forward Deep Neural Network (FDNN). In the MTL architecture, shown in figure 2, the tasks of stress recognition and user identification are unified into a single training procedure. The FDNN consists of a fully connected layer with 100 hidden units, a normalization layer, and a dropout layer. The initial fully connected layer has the role of a size balancer, meaning that it has to bring both inputs from AR

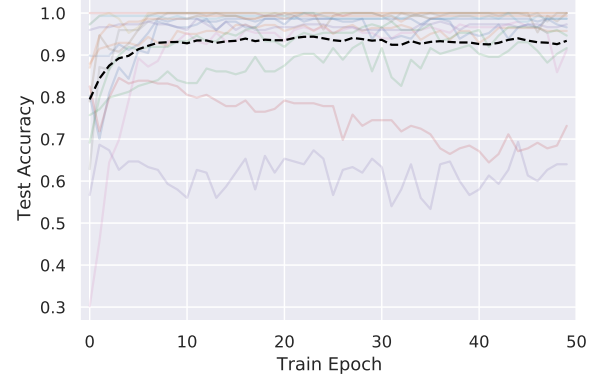
Table 2: BVP features

Features	Explanation
<i>bpm</i>	beats per minute
<i>ibi</i>	interbeat interval
<i>senn</i>	std. dev. of intervals between adjacent beats
<i>sdsd</i>	std. dev. of differences in adjacent R-R intervals
<i>rmssd</i>	root mean sq. of differences in adjacent R-R intervals
<i>pnn20</i>	ratio of differences in R-R intervals > 20ms
<i>pnn50</i>	ratio of differences in R-R intervals > 50ms
<i>hr_mad</i>	median absolute deviation
<i>sd1, sd2, S, $\frac{sd1}{sd2}$</i>	Poincaré analysis
<i>breathing_rate</i>	breaths per minute
<i>lf</i>	low frequency component
<i>hf</i>	high frequency component
<i>lf/hf</i>	low/high frequency ratio
<i>median</i>	value separating the higher half from the lower half
<i>mean</i>	arithmetic mean
<i>std</i>	standard deviation
<i>var</i>	statistical variance
<i>slope</i>	slope of the regression line
<i>min</i>	minimum value of the signal
<i>max</i>	maximum value of the signal
<i>fdmean</i>	mean value of the gradient on the BVP signal
<i>fdstd</i>	standard deviation value of the gradient on the signal
<i>drange</i>	ratio between the largest and smallest values


Figure 2: Architectural Overview of the MTL pipeline

and UR down to a commonly recognized size so that later they can be fed into a shared layer. The normalization layer is used as a model-optimization technique where the output of all the neurons is normalized [26]. In addition, it allows for a higher learning rate value to be used, thus improving the performance at which the network can learn [8]. The dropout layer, configured with a dropout rate of 0.2, is utilized to reduce the risk of model overfitting. In the MTL configuration, a shared dense layer is used to unify the outputs of the two branches. We used the *ReLU* activation function in the fully connected layers. We trained the MTL models with a learning rate of 10^{-3} , batch size of 15, and 50 epochs. The implementation of the MTL model is publicly available and can be accessed at [23].

We evaluated all the models using the *Leave-One-Subject-Out* (LOSO) validation procedure and the F_1 -score metric with the *macro* average strategy, in which the metrics for each label are calculated and their unweighted mean is found [28]. The *LOSO* is a user-independent, iterative approach in which the model is trained using the data of all the participants except one that is used for testing, suitable for determining the robustness of the model on data of unseen users.


Figure 3: Learning curves for the MTL architecture
Table 3: F1-score (%) MTL vs. Single task vs. Related work

WESAD-RF[27]	WESAD-AB[27]	RF	XGB	MTL
86.1	85.8	70.6	85.6	92.6

Table 4: F1-score (%) MTL vs. Single task

RF	XGB	MTL
52.0	40.0	29.7

5 RESULTS AND DISCUSSION

5.1 Stress Recognition

In Table 3 the results of the LOSO evaluation are shown. The first two columns (WESAD-RF and WESAD-AB) are results from the *Random Forest* and *Ada Boost* classifier from [27], whilst the rest of the columns represent the results achieved by the methods implemented in this study. The XGBoost algorithm performs better with respect to the RF methodology, for what concerns this study. One main difference between the related-work results and the results in this work is that WESAD-RF and WESAD-AB utilized data from a wrist device and from an ECG device, whereas the methods implemented in this work use only the data from the wrist device. This may be the reason why the WESAD-RF scored 15.5 percentage points more than the RF implemented in this work achieved. On the other hand, the XGB model (using only wrist data) performed similarly to the related work models, which indicates that the pre-processing pipeline and the extracted features are informative enough to replicate the results of the related work. Furthermore, from Table 3 it can be seen that the MTL approach outperformed the best related-work model and the best baseline models implemented in this study as well. To investigate the MTL model behavior for various test users, we present the learning curves from the LOSO evaluation in Figure 3. The x-axis presents the training epoch, the y-axis presents the test accuracy for the given epoch, the color-coded full lines represent the test accuracy for various test users, and the dashed black line represents the average test accuracy. From

the figure, it can be seen that there are two test users for which the MTL model performed quite worse compared to the average performance. For the rest of the users, the learning curves are quite stable, signifying a stable MTL model.

5.2 User-recognition

To evaluate the accuracy of the user-identification branch of the MTL network, we need data of the same user both in the train and in the test set, which is not plausible using LOSO evaluation. For that reason, we implemented an additional split of the data. To avoid temporal overlap between the train and the test data, we took the first 5 minutes and the last 2.5 minutes of both stress and no-stress data. Experiments were performed using LOSO approach again, but one subset of the test user was added to the train data (i.e., the first 5 minutes), and the other subset (i.e., the last 2.5 minutes) was kept for testing. The results of these experiments are presented in Table 4. While the MTL performed worse than the single-task models in this evaluation, the results still indicate that all of the models learn to recognize users to some extent, given that the accuracy of a majority classifier (a classifier that always predicts the most frequent user-id) would be 6.67% (1 out of 15 users).

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